BTS–VIII(R/S)–04–24–2284 Reg. No.

A

B.Tech. Degree VIII Semester Regular/Supplementary Examination April 2024

ME 19-205-0802 MATERIALS MANAGEMENT

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the scope, objectives and phases in materials management.
CO2: Get concept of static inventory problems.
CO3: Study dynamic inventory problems under risk.
CO4: Lear about lot sizing in material requirement planning.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	Inventory is a necessary evil. Comment.	3	L2	2	1
	(b)	Compare ABC analysis with VED analysis.	3	L1	2	1
	(c)	Explain opportunity cost with an example.	3	L1	3	1
	(d)	Explain the features of static inventory problems under risk.	3	L1	3	2
	(e)	What do you meant by Dynamic Inventory problems under risk? Show the classification of inventory problems with a neat flow chart.	3	L1	3	2
	(f)	Compare inventory profile model of Q-system for ideal and real model with neat diagrams (saw tooth diagrams).	3	L1	3	2
	(g)	Mention the different lot sizing methods used in MRP.	3	L1	3	2
	(h)	Compare dependent and independent demand systems in inventory management.	3	L1	3	2

PART B

$(4 \times 12 = 48)$

II. A company produces a mix of high technology products for use in hospitals. The annual sales data are given in the table below. Classify the items based on ABC analysis. 12 L3 1 1,2

Items	Unit price (₹)	No of units consumed	Items	Unit price (₹)	No of units consumed
P	5	1000	U	1	150
Q	10	10	V	8	1500
R	7	5	W	6	10000
S	750	100	X	3	20
T	5	2000	Y	4	9000

OR

(P.T.O.)

			Marks	BL	CO	PO
III.	(a)	Explain the importance of inventory management. Also explain inventory costs in detail.	6	L1	1	1
	(b)	A company buys in lots of 500 boxes and the delivery is happening in every 3 months in an year. The cost per box is ₹125 and the ordering cost is ₹150. The carrying cost is estimated as 20% of unit value. Find: (i) EOQ (ii) Evaluate the impact of EOQ on the total cost.	6	L3	3	1,2
IV.	(a)	ABC company requires special cycloidal gears at the rate of 300 numbers per year. Each gear costs ₹36. The procurement cost is estimated as ₹30 and the interest rates are 20%. The supplier offers a discount of ₹2 per gear for an order of 200 or above. Will it be advisable to go for discounts?	10	L3	2	1,2
	(b)	What does a high cost of risk imply?	2	L2	2	1
		OR				
V.		A large department store has purchased a new central air conditioning unit. The life time of the AC is estimated as 12 years. The manager is supposed to formulate the purchasing strategy based on the given information. If he purchases along with the unit, he can get each compressor at the rate of ₹100. If he is getting it after the failure, he needs to spend ₹1000 for the replacement. Ignore all other costs like maintenance fee etc. The table given below gives the probability distribution of the failure rates of compressors. Find the expected total cost value and the expected opportunity cost value. Suggest a suitable strategy to ensure the continuous operation of the unit.	12	L3	2	1,2

No. of failures	Probability
0	0.30
1	0.40
2	0.25
3	0.05

(Continued)

		Marks	BL	CO	PO
VI. (a)	Differentiate between P systems and Q systems with neat sketches (saw tooth diagrams).	4	L1	3	1
(b)	Weekly demand is normally distributed with a mean of 20 units and a standard deviation of 4. What is the optimum reorder point if holding costs are ₹5.00 per unit per year, the backorder cost is ₹10.00 per outage? The order quantity is 26 units and the lead time is 1 week.	4	L3	3	1
(c)	What is the optimal reorder point for the inventory problem specified below and is given in table?	4	L3	3	1,2

Annual Demand = 1800
Ordering cost = 30/order
Carrying cost = 15% of unit cost per year
Cost per unit = ₹2
Shortage cost = ₹1 per unit
A = ₹1.00 per unit backordered

Lead time demand M	Probability P(M)	Probability of stock out P(M>B)
54	0.20	0.18
55	0.07	0.11
56	0.06	0.05
57	0.03	0.02
58	0.02	0.00

OR

		Marks	BL	CO	PO
VII.	What is the difference between Approximate analysis and Exact analysis in dynamic inventory problems? For an item with weekly demand normally distributed with mean of 50 and standard deviation of $\sigma = 5$, Lead time between the order and delivery is 3 weeks. Perform Approximate Analysis based on the given cost data:	12	L3	3	1,2

Ordering cost = ₹10
Carrying cost = 12% per year per unit.
Unit cost = ₹5.

		Marks	BL	CO	PO
VIII. (a)	Explain the importance of MRP in modern production system.	4	L2	4	1
(b)	What are the inputs required by MRP and outputs generated by MRP? Explain with a block diagram.	8	L1	4	1

OR

		Marks	BL	CO	PO
IX.	XYZ company has to make a MRP methodology to schedule production for nine weeks. Refer the information given below. Final product A requires 1 unit of B and 1 unit of C. 1 unit of D is required for making 1 unit of C.	12	L3	4	1,2

PART	LEAD TIME	LOT SIZE	INVENTORY ON HAND
A	1	LFT	125
B	3	100	875
C	2	LFT	55
D	1	50	900

Master production schedule for 9 days is given below:

Day	1	4	6	7	9
Demand	100	500	500	100	95

Blooms's Taxonomy Levels
L1 – 32%, L2 – 10%, L3 – 58%.

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